

Dengjia Zhang

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Education

Johns Hopkins University

MS in Computer Science

Sept 2024 - May 2026

- GPA: 4.0/4.0

Beijing Jiaotong University

BEng in Computer Science

Sept 2020 - June 2024

- GPA: 3.84/4.0

Research Experience

Fine-Grained Multi-Modality Models

June 2025 - Dec 2025

Advisor: Reno Kriz

- Proposed a distribution-based modeling approach that maps learned tokens to Gaussian probability outputs for smoother, calibrated predictions.
- Developed a three-stage training pipeline that independently trains modality-specific encoders for efficient and effective model distillation.

Self Evolving LLM Agent via Uncertainty-aware Rewards

Aug 2025 - Nov 2025

Advisor: Kenton Murray and Hua Wei

- Proposed a unified uncertainty framework that integrates multiple token-level uncertainty measures, enabling a more accurate characterization of model confidence and providing dense, informative supervision derived directly from these confidence estimates.
- Introduced a failure-aware reward reshaping mechanism that integrates these signals into both step- and trajectory-level rewards, improving exploration efficiency and stability.

DPO for Stylistic Calibration of LLM Confidence

May 2025 - June 2025

Advisor: Kenton Murray

- Developed a novel RLHF-based framework that calibrates model confidence through linguistic style.
- Formulated a style-aware DPO objective to align expressed confidence with factual correctness.
- Revealed that RLHF tunes expressive style, not intrinsic belief calibration.

Few-Shot Semantic Segmentation Method

Feb 2023 - July 2023

Advisor: Runmin Cong

- The main goal is to develop a more precise semantic segmentation model through small sample sizes.
- Proposed top-k Cyc-Consistent model to help model integrate the information in small size of models.

Physiological Signal Classification Based on Confidence Calibration

Oct 2022 - Apr 2023

Advisor: Jing Wang

- Experimented with U-Sleep model to improve the performance of sleep signal classification.
- Developed Gaussian Sampling to improve the robustness of the model and used Focal Loss to address the gross imbalance between positive and negative samples.

Publications

Seeing Through the MiRAGE: Evaluating Multimodal Retrieval Augmented Generation

Oct 2025

Alexander Martin, William Walden*, Reno Kriz*, **Dengjia Zhang**, Kate Sanders, Eugene Yang, Chihsheng Jin, Benjamin Van Durme

<https://arxiv.org/abs/2510.24870> 

Projects

Generating and Reranking Diverse Answers in Video Question Answering*Aug 2025 - Sept 2025*
[Trec-VQA\(short paper\)](#) [🔗](#)

- Proposed a generate and rerank setting for producing diverse ranked lists in open-ended Video QA.
- Introduced Masked Pointer Cross-Entropy Loss with Rank Weights, explicitly incorporating ranking information into the optimization process.

A Fairer Evaluation for the Time Awareness in Instruction-Tuned LLMs*Mar 2025 - Apr 2025*
[Evaluation](#) [🔗](#)

- Developed a question-answering-based framework to fairly evaluate temporal reasoning in instruction-tuned large language models.
- Designed and implemented two complementary tasks—precise date prediction and time-conditioned question answering.

3D Face Style Editing*Jan 2024 - June 2024*
[3dface-editting\(project\)](#) [🔗](#)

- Achieved conversion from a single 2D face image to a 3D face model, retaining the key features of the original image
- Obtained renderings of the 3D face model from different perspectives.
- Introduced StyleClip, allowing people to customize facial expressions and styles.

Technologies

Languages: C++, C, Java, Python, SQL

Service

<i>WACV Reviewer</i>	<i>2026</i>
<i>MAGMaR Workshop Reviewer</i>	<i>2026</i>
<i>WACV Reviewer</i>	<i>2025</i>
<i>MAGMaR Workshop Reviewer</i>	<i>2025</i>